Gabriel Mastrilli

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Education and Academic Background

2023–2026: PhD thesis supervised by Frédéric Lavancier (ENSAI, Rennes) and Bartłomiej Błaszczyszyn (INRIA/ENS Paris): *Spectral Inference for Spatial Point Processes.*

2022–2023: Master's degree (M2) in Mathematics, Rennes.

- Probability track: Stochastic Processes, Stochastic Calculus, Dynamical Systems and Ergodic Theory, Statistics of Stochastic Processes, Parametric Estimation, Non-parametric Estimation, Zeros of Gaussian Processes, Mean Field Stochastic Differential Equations.
- Analysis track: Spectral Theory, Sobolev Spaces and Elliptic Equations, Hyperbolic Equations, Finite Element Method, Numerical Methods for Transport Equations, Birkhoff Normal Form for Hamiltonian PDEs.

2021–2022: Gap year.

2020–2021: Preparation for the French *agrégation* (Mathematics), as part of the ENS Rennes Mathematics Magistère program.
Option: Probability and Statistics. Ranked 16th.

2019–2020: M1 (first-year Master's degree) in Pure and Applied Mathematics, ENS Rennes Mathematics Magistère program.
Awarded an excellence scholarship from ENS Rennes for 2020–2021 and 2022–2023.

2018–2019: L3 (Bachelor's level, third year) in Mathematics, Research track, ENS Rennes Mathematics Magistère program.
Awarded a Master's scholarship from the Henri Lebesgue Center for 2019–2020.

2017–2018: First year at ISAE-SUPAERO, Toulouse, alongside the L3 Mathematics program at Paul Sabatier University, Toulouse.

2015–2017: Preparatory classes (CPGE), MPSI then MP*, Lycée Faidherbe, Lille.

2015: Scientific Baccalauréat, Lycée Jean-Joly, Réunion island.

Teaching Experience

2023–2026: Teaching Assistant at ENS Rennes.

- Exercise sessions + Python sessions in Statistics for Master 1 Mathematics (22h+3h/year).
- Exercise sessions + Python sessions in Statistics for Master 1 Mechatronics (2h+2h/year).
- Supervision of two directed readings for L3 students.
- Probability course supplement for *agrégation* preparation (6h/year).
- Member of the mock oral exams jury for the probability option of the *agrégation* (20h/year).

2022–2023: Oral exams ("colles") at Lycée Chateaubriand:

- MP* (1h/week).
- PC* (1h/week).

2022-2023: Teaching assistant at INSA Rennes:

- Analysis for 3rd-year engineering students (10h).
- Linear Algebra and ODEs for 2nd-year engineering students (36h).

Publications

- G. Mastrilli, *Minimax Estimation of the Structure Factor of Spatial Point Process*, 2025, preprint arXiv:2511.14551.
- G. Mastrilli, *Asymptotic Fluctuations of Smooth Linear Statistics of Independently Perturbed Lattices*, 2025, *Journal of Statistical Physics.*
- G. Mastrilli, B. Błaszczyszyn, F. Lavancier, *Estimating the Hyperuniformity Exponent of Point Processes*, 2024, to appear in *Bernoulli.*
- D. Hawat, G. Mastrilli, R. Bardenet, R. Lachièze-Rey, *Repelled point processes with application to numerical integration*, 2023, to appear in *Scandinavian Journal of Statistics.*

Talks

Invited Conference

- June 2025. GéoSto 2025, Grenoble. *Estimating the Hyperuniformity Exponent of Spatial Point Processes.*

Seminars

- February 2025. Probability and Statistics Seminar, Poitiers. *Estimating the Hyperuniformity Exponent of Spatial Point Processes.*
- December 2024. PhD Students' Statistics Seminar, Rennes. *Statistical Inference for Spectral Properties of (Euclidean and Spherical) Spatial Point Processes.*
- November 2024. "Point Processes and Applications" Working Group, Lille. *Estimating the Hyperuniformity Exponent of Spatial Point Processes.*
- September 2024. Dyogene Group Seminar, INRIA Paris. *Estimating the Hyperuniformity Exponent of Spatial Point Processes.*

Contributed Conferences

- June 2026. 23rd Workshop on Stochastic Geometry, Stereology and Image Analysis, Split. *Asymptotic Fluctuations of Smooth Linear Statistics of Independently Perturbed Lattices.*
- May 2026. 11e Rencontres des Jeunes Statisticien-ne-s, Porquerolles. *Whittle estimation for point processes.*
- April 2026. 3e Doctoral Students' Day in Statistics, Rennes. *Whittle estimation for point processes.*
- October 2025. Random Geometry Graph and Extremes, Dubrovnik. *Asymptotic Fluctuations of Smooth Linear Statistics of Independently Perturbed Lattices.*
- July 2025. SPA 25, Wrocław. *Minimax Estimation of the Structure Factor of Spatial Point Processes.*
- June 2025. SSIAB 16, Smögen. *Minimax Estimation of the Structure Factor of Spatial Point Processes.*
- May 2025. 2nd Doctoral Students' Day in Statistics, Rennes. *Estimating the Spectral Properties of Spatial Point Processes.*
- October 2024. PEPPER Workshop, Münster. *Estimating the Hyperuniformity Exponent of Spatial Point Processes.*
- June 2023. SGSIA 22, Bad Herrenalb. *Estimating the Hyperuniformity Exponent of Spatial Point Processes.*

Reading Groups

- March 2025. Dyogene/MathNet Reading Group, INRIA Paris. *"Introduction to Minimax Theory for Spatial Point Processes: First- and Second-Order Estimation.*
- February 2023. Dyogene/MathNet Reading Group, INRIA Paris. *Estimation of the Hyperuniformity Class.*

Internships and Research Seminars

- March–June 2023: Master's thesis supervised by Frédéric Lavancier (ENSAI, Rennes) and Bartłomiej Błaszczyszyn (INRIA/ENS Paris).
  Topic: Statistical inference for hyperuniform point processes.
- October–December 2022: Research Seminar in the M2 Mathematics Program, Rennes. Supervised by François Bolley and Paul-Eric Chaudru de Raynal.
  Topic: Numerical approximation of scalar reflected SDEs via Mean Field SDEs.
- April–August 2020 (COVID period): Statistical Physics Internship (remote), supervised by Gabriel Lemarié and Keith Slevin. Continuation of the 2018 internship.
- March–April 2020 (COVID period): Internship (remote) supervised by Magalie Fromont. Topic: Non-parametric Statistical Estimation, based on *Introduction to Nonparametric Estimation*, Tsybakov.
- May–July 2019: Internship supervised by Andrew Duncan (Imperial College London). Topic: Application of PDMPs (Piecewise Deterministic Markov Processes) to Monte Carlo methods.
- January–May 2019: Directed Reading supervised by Vincent Duchêne (Rennes).
  Topic: Concentration-compactness methods applied to the stability of nonlinear hyperbolic PDEs.
- June–July 2018: Statistical Physics Internship supervised by Gabriel Lemarié (Toulouse).
  Topic: Anderson Localization. Numerical study of phase transitions in random graphs.